Minseo Kim

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Research Interests

AI Systems, Efficient Inference across Modern Architectures and Workloads

Education

- **University of California, Berkeley** — *Aug 2026 – Present*
  Ph.D. in Computer Science — Berkeley, CA, USA
  ◦ Advisor: Prof. Kurt Keutzer
- **Seoul National University** — *Mar 2020 – Aug 2026*
  *B.S. in Computer Science and Engineering, **Summa cum laude** (GPA: 4.03/4.30)* — Seoul, South Korea
  ◦ Military service in the Republic of Korea Air Force (Oct 2021 – Jul 2023)
- **University of California, Berkeley** — *Jun 2025 – Aug 2025*
  Summer Session Visiting Student (GPA: 4.0/4.0) — Berkeley, CA, USA
- **Korea Minjok Leadership Academy** — *Mar 2017 – Feb 2020*
  *Secondary Education, **First-class honors** in STEM (1st / 73)* — Gangwon-do, South Korea

Research Experience

- **Berkeley Artificial Intelligence Research (BAIR) Lab [🌐]** — *Aug 2026 – Present*
  Ph.D. Student (Advisor: Prof. Kurt Keutzer) — Berkeley, CA, USA
  ◦ Efficient inference in hybrid attention and looped models; agent harness optimization.
- **Berkeley Artificial Intelligence Research (BAIR) Lab [🌐]** — *Jun 2025 – Jan 2026*
  Undergraduate Visiting Researcher (Advisor: Prof. Kurt Keutzer) — Berkeley, CA, USA
  ◦ LLM inference on compute-in-memory (CIM) hardware; diffusion LLM inference.
- **SNU Architecture and Code Optimization Lab [🌐]** — *Jan 2025 – Feb 2025*
  Undergraduate Researcher (Advisor: Prof. Jae W. Lee) — Seoul, South Korea
  ◦ Storage I/O optimization for approximate nearest neighbor search (ANNS) systems.

Work Experience

- **FuriosaAI [🌐]** — *Jan 2026 – Jul 2026*
  AI Algorithm Research Intern — Seoul, South Korea
  ◦ Efficient post-training (RL, on-policy distillation); agent systems (data synthesis, harness optimization).
- **Deeping Source Inc. [🌐]** — *Jul 2024 – Aug 2024*
  ML Engineering Intern — Seoul, South Korea
  ◦ Pedestrian attribute recognition model development with gaze estimation.

Publications

* Equal contribution

[1] Minseo Kim*, Minjae Lee*, Seunghyuk Oh, Kevin Galim, Donghoon Kim, Coleman Hooper, Harman Singh, Amir Gholami, Hyung Il Koo, Wonjun Kang (2026). **EfficientRollout: System-Aware Self-Speculative Decoding for RL Rollouts**. In Proceedings of *NeurIPS 2026*; *ICML 2026 AdaptFM Workshop*. [Paper] [Code]

[2] Wonjun Kang*, Kevin Galim*, Seunghyuk Oh, Minjun Kang, Sanghyun Park, Donghoon Kim, Minjae Lee, Minseo Kim, Rishabh Tiwari, Yuchen Zeng, Hyung Il Koo, Kangwook Lee (2026). **AsyncOPD: How Stale Can On-Policy Distillation Be?** In Proceedings of *NeurIPS 2026*; *ICML 2026 DEMO Workshop*. [Paper] [Code]

[3] Omin Kwon, Yeonjae Kim, Doyeon Kim, Minseo Kim, Yeonhong Park, Jae W. Lee (2026). **MAGE: All-[MASK] Block Already Knows Where to Look in Block Diffusion LLM**. In Proceedings of *NeurIPS 2026*; *ICML 2026 AdaptFM Workshop* (Best Paper Award). [Paper]

[4] Minseo Kim, Chenfeng Xu, Coleman Hooper, Harman Singh, Ben Athiwaratkun, Ce Zhang, Kurt Keutzer, Amir Gholami (2026). **CDLM: Consistency Diffusion Language Models for Faster Sampling**. In Proceedings of *MLSys 2026*. [Paper] [Code]

[5] Taebaek Hwang*, Minseo Kim*, Gisang Lee, Seonuk Kim, Hyunjun Eun (2025). **KRETA: A Benchmark for Korean Reading and Reasoning in Text-Rich VQA Attuned to Diverse Visual Contexts**. In Proceedings of *EMNLP 2025*. [Paper] [Project] [Dataset]

[6] Bonggeun Sim, Yushin Kim, Minseo Kim, Yeonhong Park, Jae W. Lee (2025). **InstANNS: Scalable Approximate Nearest Neighbor Search via Cost-Efficient In-Storage Processing**. In Proceedings of *CIKM 2025*. [Paper]

Preprints & Workshop Papers

* Equal contribution; † Co-corresponding authors

[1] Younghyun Kim*, Wooseok Seo*, Minseo Kim, Seunghyuk Oh, Minjae Lee, Donghoon Kim, Hyung Il Koo, Jaehyung Kim†, Wonjun Kang† (2026). **Elicitation or Expansion? Demystifying Gains from Agent Harness Optimization**. Conference submission under review.

[2] Omin Kwon, JoongWon Shin, Minseo Kim, Kurt Keutzer, Sehoon Kim, Jae W. Lee (2026). **SketchSSM: Write to the Full State, Read from a Compact Sketch**. *NeurIPS 2026 LCFM Workshop*. Conference submission under review. [Paper]

[3] Sebastian Zhao*, Minseo Kim*, Coleman Hooper*, Luca Manolache, Michael W. Mahoney, Yakun Sophia Shao, Kurt Keutzer, Amir Gholami (2026). **LLM Inference in a Flash!** *ISCA 2026 MLArchSys Workshop* (Oral). [Paper]

[4] Seunghyuk Oh, Minjae Lee, Kevin Galim, Minseo Kim, Hyung Il Koo, Wonjun Kang, Hanbaek Lyu, Kangwook Lee (2026). **CATS: Inference-aligned SFT for Diffusion LLMs via Context-sensitivity Aware Trajectory Sampling**. *ICLR 2026 DeLTa Workshop*. [Paper]

[5] Minseo Kim, Coleman Hooper, Aditya Tomar, Chenfeng Xu, Mehrdad Farajtabar, Michael W. Mahoney, Kurt Keutzer, Amir Gholami (2025). **Beyond Next-Token Prediction: A Performance Characterization of Diffusion versus Autoregressive Language Models**. *arXiv preprint*. [Paper]

Projects

- **Open-Source Contribution: Samsung NNTrainer** [] *2025*
  - Gradient checkpointing and optimizers (Lion, Sophia) for the on-device training stack.
- **ARC-AGI Challenge Solver** [] *2025*
  - Placed **1st among 35 teams** in a Deep Learning course project. [Technical report]
- **Art College Graduation Exhibition Sales Website** [] *2024*
  - Backend team; collaborated with the Metal Craft and Ceramics departments at SNU.
- **Windows Software Vulnerability Research & Exploitation** *2023*
  - Disclosed **three CVE-registered vulnerabilities** (CVE-2024-11510, CVE-2024-11511, CVE-2024-11512).

Honors and Awards

- **2025 AI Chip Contest – Grand Prize (Top Award), Korea** *Dec 2025*
  - AI semiconductor application track; optimized vision preprocessing on NPUs (KRW 10M prize).
- **Gaheonsindo Foundation Scholarship (Full Undergraduate Tuition)** *2024 – 2026*
- **SNU Specialized Semiconductor College Scholarship** *2024 – 2026*
- **Ministry of Science and ICT Award - Whitehat School Outstanding Graduate** *Mar 2024*
  - Ranked 1st out of 309 participants (KRW 3M prize).
- **Merit-based Scholarship** *2021, 2023*

Leadership & Talks

- **Research Seminar, Cerebras** *Jan 2026*
  - Co-presented a one-hour online seminar on diffusion language models.
- **President, Guardian (SNU CSE Security Research Club)** *Aug 2024 – Aug 2025*
  - Organized and delivered weekly seminars for new members on system hacking, reversing, web security, and cryptography.
- **Presenter (28th Winter Hacking Camp) – Accepted Talk** *Feb 2024*
  - Delivered a 1-hour instructional session on Windows software vulnerability detection and analysis.

Language Proficiency

Korean (Native), English (Fluent)